Karan Gupta

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Education

University of Waterloo — Expected June 2030
Bachelor of Applied Science in Computer Engineering, Minor in Economics — ***GPA**: 3.95*

Experience

Software Development Engineer Intern — May 2026 – Present
Amazon — *Toronto, ON*

- Reduced legacy service migration time by 98% (2 weeks to under 2 hours) by designing an adaptive AI discovery agent and deterministic code-generation workflows, authoring 20 agent skills to translate Java monoliths
- Automated the generation of over 13,000 lines of production code to transform a 9,400-line warehouse scheduling monolith into a pluggable microservice architecture deployed across every fulfillment center
- Re-architected the core decision engine into stateless solvers on AWS ECS, using Java CompletableFutures to orchestrate concurrent S3 data extractions, preventing execution bottlenecks during an 8-minute scheduling cron
- Built a semantic-diff validation harness to replay production JSON payloads against generated endpoints, evaluating behavioral parity via a confidence metric that flags inconclusive noise across 42 execution paths

AI Software Engineer, Evaluation Lead — May 2026 – Present
Wat.ai (TRACE Subteam) — *Waterloo, ON*

- Implemented multi-step execution tracing and failure isolation for TRACE, an AI agent reliability engine, reducing root-cause diagnostic latency by 70% for complex RAG and tool-use workflows
- Eliminated 95% of runtime non-determinism across 4 downstream production applications by implementing a canonical verifier interface and Pydantic schemas to enforce strict data contracts over LLM outputs
- Deployed a high-throughput, 7-check deterministic execution engine that intercepts critical policy violations to enforce 100% compliance for financial-action workflows

Software Engineering Intern — January 2026 – April 2026
Manulife Financial Corporation — *Waterloo, ON*

- Reduced system downtime risk by scripting a New Relic data exporter and architecting NRQL queries, enabling early detection of 3+ resource bottlenecks before system failure
- Designed a Python analysis script and SQL telemetry pipeline to process clickstream data, identifying 20+ suspicious user anomalies by applying mathematical modeling and cross-referencing behavioral logs
- Implemented 15+ Salesforce features to automate data entry during document uploads for specialized lending workflows, reducing manual input for high-value client processing using Apex and LWC

Autonomy Software Developer — September 2025 – February 2026
Waterloo Aerial Robotics Group — *Waterloo, ON*

- Improved real-time autonomous UAV signal detection accuracy by 13% by developing OpenCV computer vision algorithms, optimizing performance through iterative parameter tuning and extensive flight simulation validation
- Built multi-process telemetry and command systems in Python using PyMAVLink to simulate flight communication, reducing message latency by 30% across distributed processes

Projects

Reparo (Hack Canada 2026 Winner) | *Python, Gemini API, React, Node.js, Swift, PyTorch*

- Classified products and assessed repairability with 90%+ accuracy via a Gemini vision-powered agentic AI
- Reduced user search time for replacement parts by 70% and enabled access to 1,000+ real-time listings by integrating Shopify Storefront API and SerpAPI to automate matching and streamline repair-to-checkout flow
- Won 1st place in Reactiv Track ($5,000) at Hack Canada 2026; shortlisted for Most Complex AI Hack

DeliriumWatch | *Raspberry Pi, Arduino, Python, TensorFlow, scikit-learn, OpenCV, HTML, CSS, Flask, C/C++*

- Built a real time Python monitoring pipeline with SQL-backed secure login and role-based access control for hospital staff, resulting in over 90% reduction in manual environmental monitoring during simulated testing
- Developed a real-time computer vision pipeline integrating OpenCV with a custom-trained machine learning model (TensorFlow, scikit-learn) for continuous eye detection, surfacing patient anomalies to a live Flask dashboard to reduce average nurse response time by 40% during simulated testing

Skills

Languages: Python, Java, C, C++, Swift, TypeScript, JavaScript, SQL, HTML, CSS
Frameworks & Libraries: PyTorch, TensorFlow, React, Node.js, FastAPI, OpenCV, PyMAVLink, JUnit, Mockito
Developer Tools: AWS (CDK, ECS, Lambda, S3), Git, Docker, Kubernetes, Raspberry Pi, Arduino